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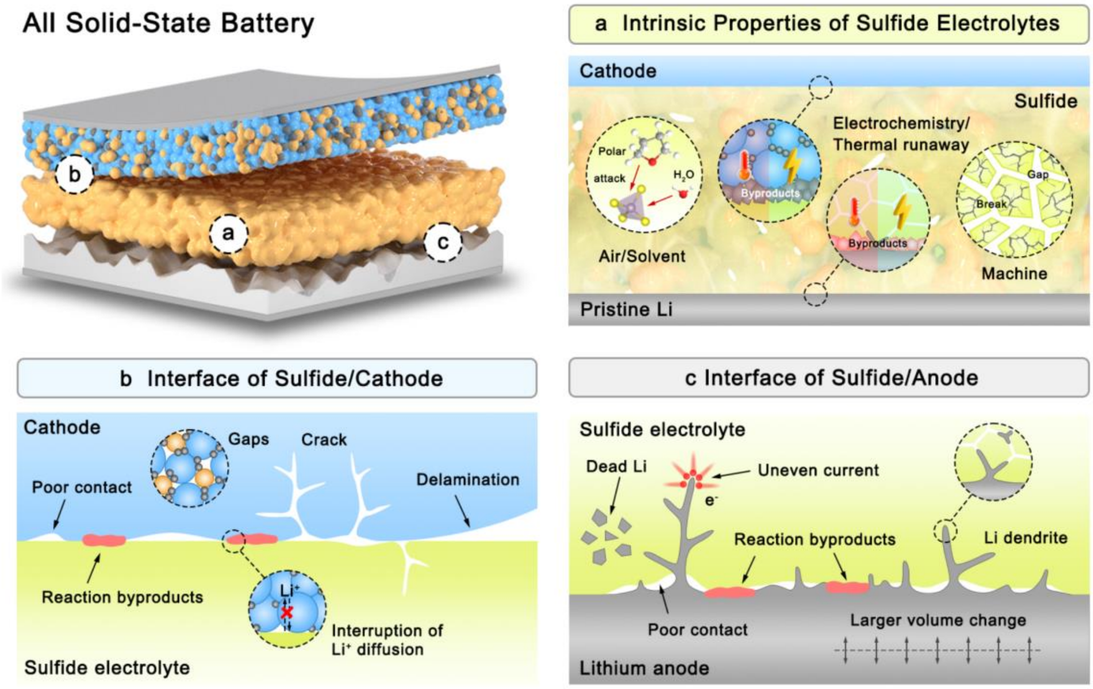

Figure 1. Schematic illustration of the key challenges faced by sulfide-based ASSBs at both the electrolyte intrinsic properties and electrode interfaces.

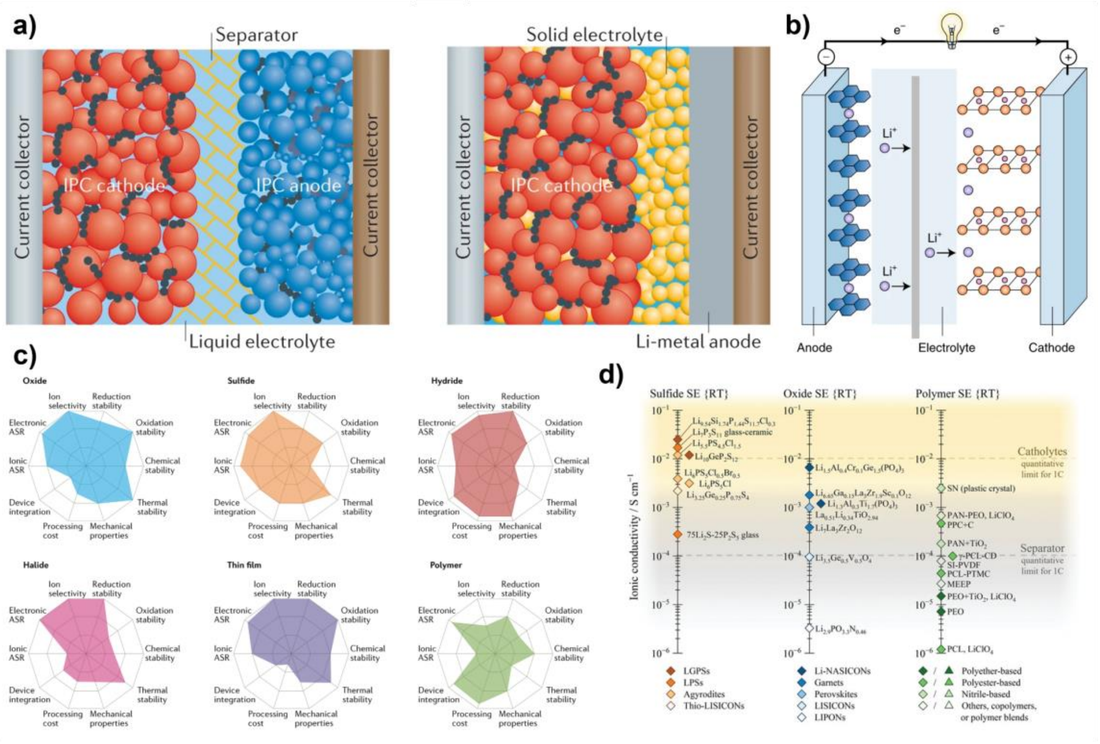

Figure 2. a) Schematic diagrams of liquid LIBs and ASSBs [1]. b) The operating principle of conventional lithium batteries [42]. c) Comparative analysis of performance characteristics among various SEs and d) ionic conductivity of representative SEs [50,57].

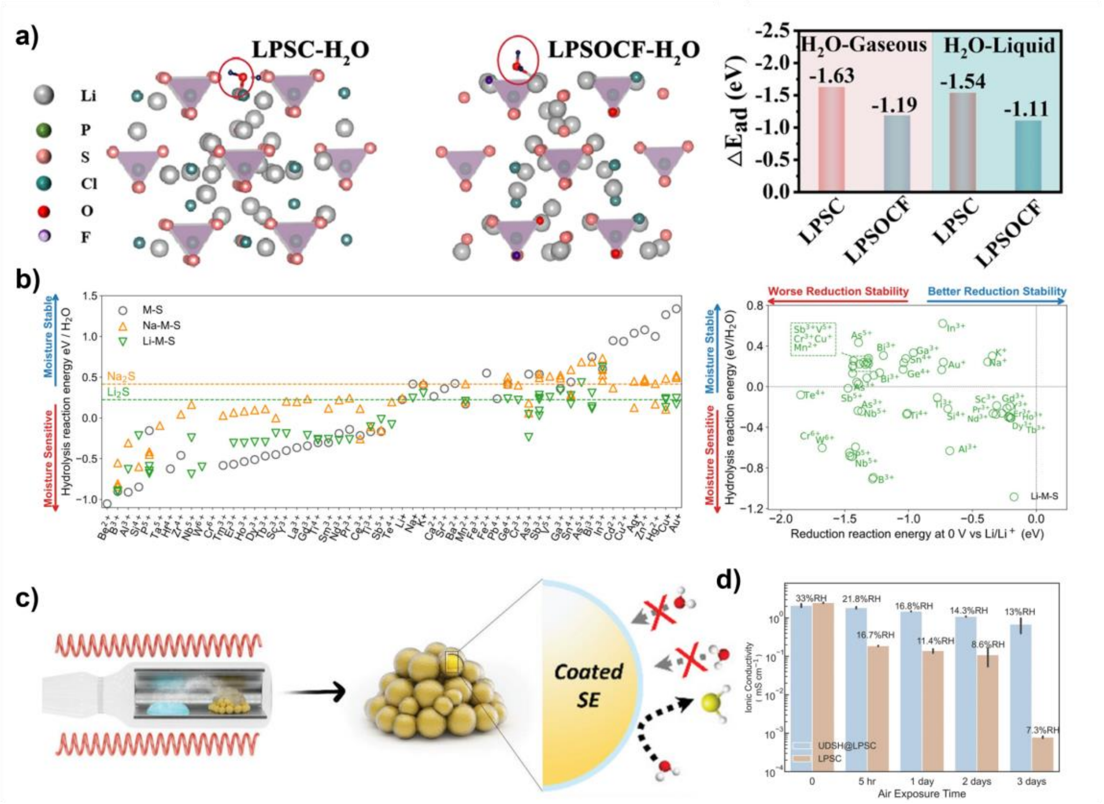

Figure 3. a) Adsorption configurations and corresponding adsorption energies of H_2O on the surfaces of sulfide SEs [81]. b) Effects of different cations on the hydrolysis reaction energy of sulfide SEs and cation selectivity guidance chart [84]. c) Vapor deposition method for constructing ultrathin superhydrophobic layers on sulfide SE surfaces [87]. d) Ionic conductivity of 1-undecanethiol-coated sulfide SEs as a function of exposure time [86].

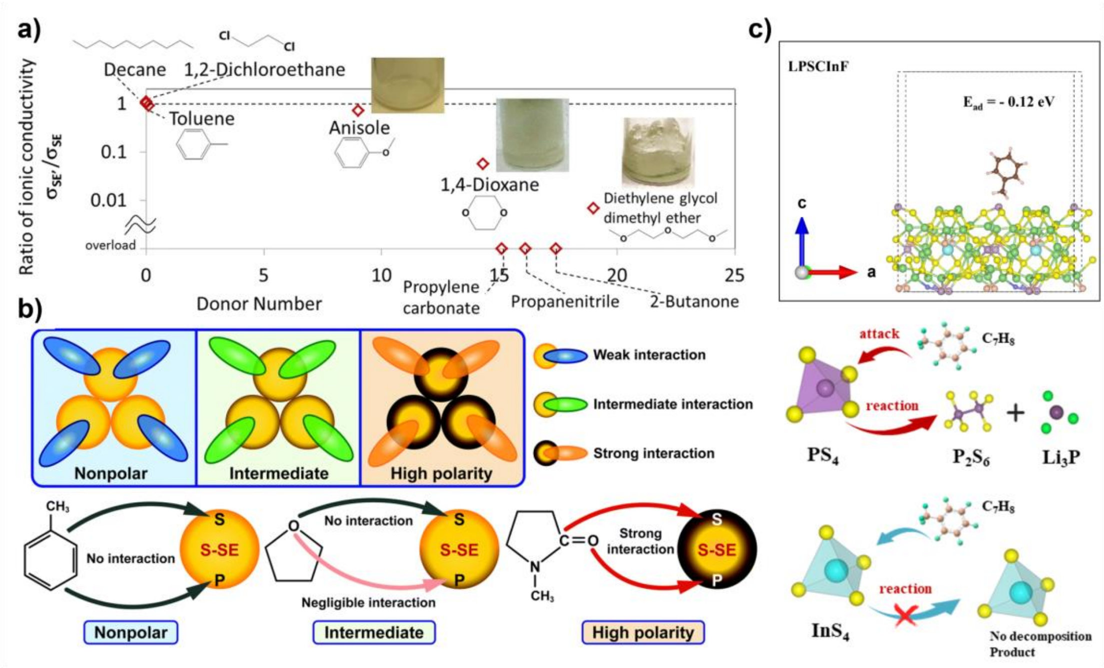

Figure 4. a) Variation in Ion Conductivity of $Li_7P_3S_{11}$ SE following exposure to different organic solvents [92]. b) Mechanism of interaction between organic solvents of different polarities and SEs [24]. c) InF_3-modified sulfide SE enhances solvent stability [97].